UNIVERSITY QUESTION BANK WITH MODEL ANSWERS
Static Inventory Problems under Risk

SECTION A – SHORT ANSWER QUESTIONS (2–3 Marks)

Q1. Define static inventory problem under risk.
A static inventory problem is a single-period inventory decision problem in which only one order is placed under uncertain demand, where probabilities of demand are known.

Q2. Why are static inventory problems called one-shot decisions?
Because once the order is placed, no further replenishment or revision is possible.

Q3. Give two examples of static inventory problems.
Newspaper vendor, Christmas tree seller.

Q4. What is excess cost?
Excess cost is the loss incurred when ordered quantity exceeds demand, leading to unsold inventory.

Q5. What is shortage cost?
Shortage cost is the loss due to unmet demand, including lost profit and goodwill.

SECTION B – DESCRIPTIVE QUESTIONS (5–6 Marks)

Q6. Explain the general characteristics of static inventory problems under risk.
Static inventory problems involve a single ordering decision, uncertain demand with known probabilities, no scope for replenishment, and costs arising from excess or shortage. The objective is to minimize expected cost.

Q7. Explain the Christmas Tree Problem.
The Christmas tree problem is a classic static inventory problem where goods have demand only for a short period. Unsold stock has little value after the season. The decision involves balancing excess and shortage costs under uncertain demand.

Q8. What is a total cost matrix?
A total cost matrix shows the cost incurred for each decision alternative under each possible demand level. Rows represent order quantities and columns represent demand states.

Q9. What is an opportunity cost matrix?
Opportunity cost matrix represents the regret associated with not choosing the best decision for a given demand. It is obtained by subtracting minimum column cost from each element.

SECTION C – LONG ANSWER QUESTIONS (8–10 Marks)

Q10. Explain static inventory problems under risk with suitable examples.
Static inventory problems under risk involve one-time ordering decisions where demand is uncertain but probabilistic. Examples include newspapers, blood banks, and seasonal fashion items. The decision aims to minimize expected cost.

Q11. Explain total cost matrix and opportunity cost matrix with explanation.
The total cost matrix evaluates cost outcomes for each decision-demand combination. Opportunity cost matrix converts these costs into regret values to apply minimax regret criterion for decision making.

Q12. Explain cost of risk in static inventory problems.
Cost of risk is the additional cost incurred due to uncertainty in demand. It is defined as the difference between expected cost under risk and cost under certainty.

Q13. Derive the mathematical formulation for discrete demand case.
Let demand take values D1, D2, … Dn with probabilities P1, P2, … Pn. Expected cost is given by:
$E(C) = \Sigma Pi \times C(Q, Di)$
The optimal order quantity minimizes expected cost.

Q14. Explain continuous demand case in static inventory problems.
When demand is continuous, expected cost is expressed as:
$E(C) = \int C(Q, D) f(D) dD$
The optimal order quantity satisfies the critical ratio:
$P(D \leq Q) = Cu / (Cu + Co)$

SECTION D – APPLICATION QUESTIONS

Q15. Explain how static inventory models are applied in hospitals.
Hospitals use static inventory models for blood banks and emergency medicines where demand is uncertain and items are perishable.

Q16. Explain application of static inventory problems in retail.
Retailers use these models for seasonal goods to balance excess stock clearance and shortage losses.